What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

H1 blockers – Antihistamines

Last updated November 4, 2025 · 9 min read · ScholarRx
Adapted by: Diptiman Bose, Trager Hintze, Daedelus Hoffman
Editors: Jordan Finn, Russell Johnson
Adapted from: Antihistamines and Decongestants

Listen to this Brick
0:00 / 16:28 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Explain the mechanisms of action of antihistamines.
2. Classify antihistamines into their respective generations based on central vs. peripheral effects.
3. Discuss the clinical uses of antihistamines based on their pharmacologic effects.
4. Differentiate the potential side effects of antihistamine drugs, based on their effects on histaminic versus muscarinic receptors.

CASE CONNECTION

SY, a 78-year-old male, is at your office to evaluate excessive sleepiness. He says, "I just can't seem to stay awake. I sit down to watch the news or read the paper, and I start to doze off. I just want to get back out to tend the garden, mow the lawn, but I'm too tired. It's a shame; everything has just started to bloom." SY takes several over-the-counter drugs to help him with "aches and pains" as well as nasal congestion that is worse when he is in the garden a lot. He can't remember the names of the drugs.

Recognizing that many things might cause this sleepiness, you start with the basics, such as reviewing his medications for recent changes, reviewing his medical history, and performing a physical exam.

What could be causing SY's new sleepiness? Consider your answer as you read, and we'll revisit SY at the end of the brick.

GO TO CONCLUSION ↓

What Are the Effects of Histamine?

Physiology of Histamine. Before looking at how antihistamines work, let's review the normal function of histamine and histamine receptors.

Histamine mediates local immune responses throughout the body. It also regulates physiologic act and acts as a neurotransmitter in the brain, spinal cord, and uterus. In this brick, we will focus on the role of histamine in regulating the immune response.

Histamine is stored in most body tissues and is primarily found in mast cells and basophils. Several events can trigger histamine release, but it is often activated in response to an **allergen** like dust, pollens, or other irritants (Figure 1, left).

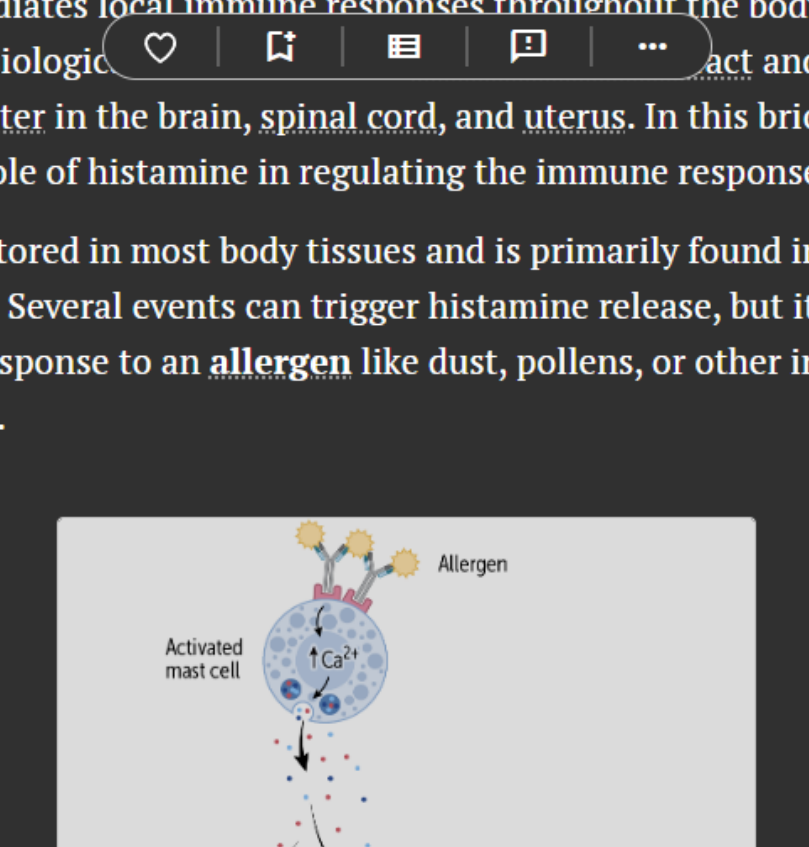


Figure 1

Once histamine is rel e receptors (H1, H2, H3, or H4). **H1 receptors** are widespread throughout the body and binding of histamine to these receptors causes various effects like vasodilation, itching, and hyper-secretion by glandular tissue, leading to nasal, chest, or ear congestion. When activated, H1 receptors in the brain promote wakefulness. In contrast, H2 histamine receptors are primarily found in gastric cells, H3 in the neurons, and H4 in the bone marrow. **H1 receptors**, located in both the central and peripheral nervous systems, are targeted in the medication therapy discussed in this brick.

How do Antihistamines Work? (LO#1)

H1 antihistamines (aka histamine antagonists and histamine inverse agonists) make up a class of drugs that acts by **competitively binding to H1 histamine receptors** on cell surfaces.

H1 antihistamines bind and antagonize the H1 receptors on endothelial cells adjacent to vascular smooth muscle cells (Figure 1, right). The binding leads to **vasoconstriction** by inhibiting the release of nitric oxide and prostacyclin. The vasoconstriction reduces edema formation, lessening runny nose symptoms. Further, the H1 antihistamines suppress the action of histamine on nerve endings, which reduces the flare and itch associated with endogenous histamine release. There are some slight mechanistic differences between first- and second-generation antihistamines, which will be discussed below.

How do H1 antihistamines reduce runny nose symptoms?

How do we classify antihistamines? (LO#2)

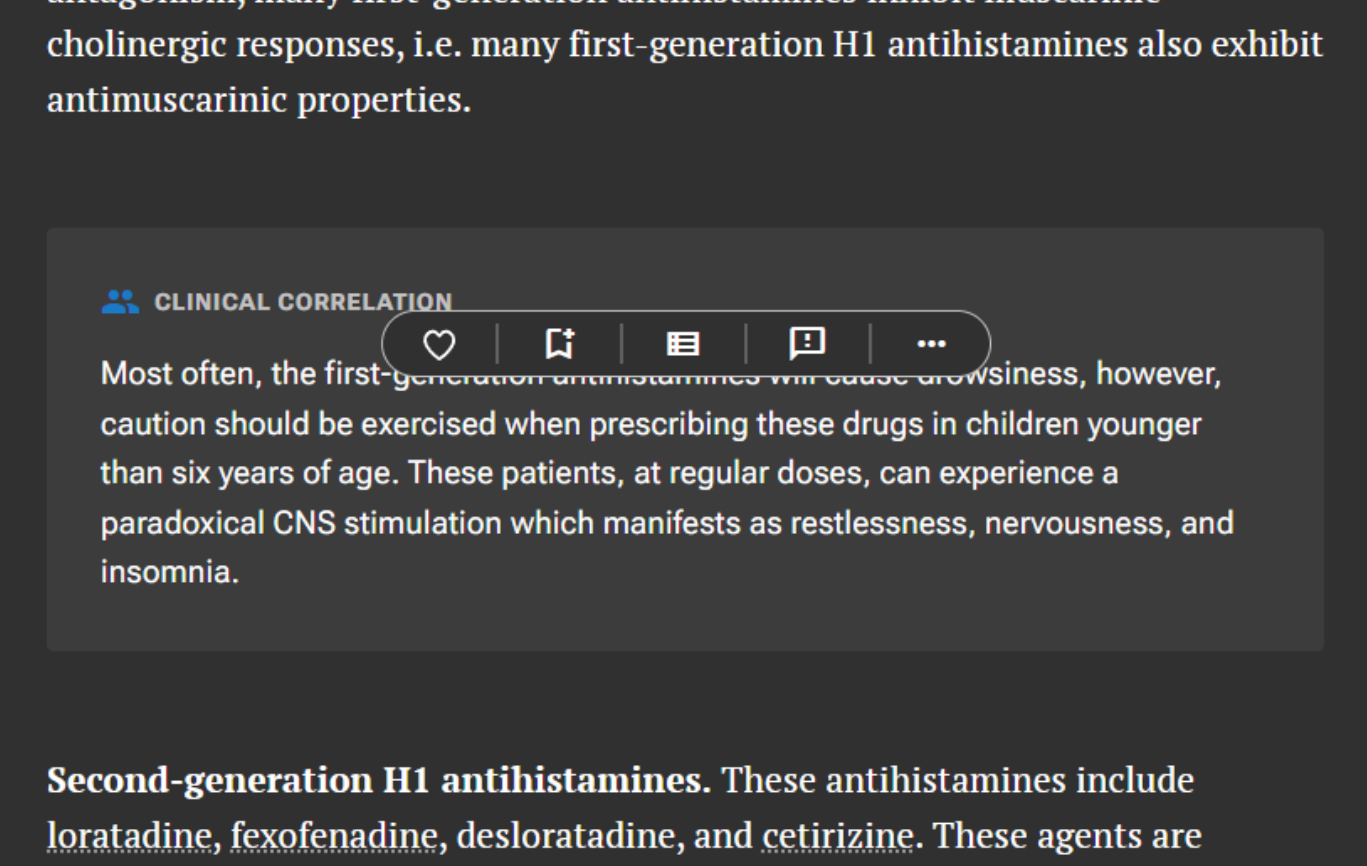
Types. There are currently two classes of H1 antihistamines: first-generation and second-generation.

First-generation H1 antihistamines. These include drugs such as diphenhydramine, chlorpheniramine, hydroxyzine, doxylamine, promethazine, and dimenhydrinate. Most are sold over the counter in the United States. These drugs are lipophilic and readily cross the blood-brain barrier. This means they block centrally located H1 receptors in the brain that usually promote wakefulness and can cause **drowsiness**. In addition to H1 histamine antagonism, many first-generation antihistamines inhibit muscarinic cholinergic responses, i.e. many first-generation H1 antihistamines also exhibit antimuscarinic properties.

CLINICAL CORRELATION

Most often, the first-generation antihistamines will cause drowsiness, however, caution should be exercised when prescribing these drugs in children younger than six years of age. These patients, at regular doses, can experience a paradoxical CNS stimulation which manifests as restlessness, nervousness, and insomnia.

Second-generation H1 antihistamines. These antihistamines include loratadine, fexofenadine, desloratadine, and cetirizine. These agents are termed *non-sedating*. Unlike first-generation antihistamines, these agents are hydrophilic, reducing their penetration of the blood-brain barrier. This means these drugs act primarily in on H1 receptors in the periphery and cause less sedation than first-generation H1 antihistamines.



Why do second-generation H1 antihistamines cause less sedation?

How are Antihistamines Used Clinically? (LO#3)

Given the widespread nature of histamine receptors throughout the body and the antimuscarinic activity caused by many first-generation H1 antihistamines, H1 antihistamines have various clinical uses.

Allergies - As H1 receptors on smooth muscle are blocked, it inhibits the release of nitric oxide and prostacyclin, resulting in vasoconstriction and reduction of edema formation. First-generation antihistamines (e.g., diphenhydramine) are also an adjunct treatment in severe allergies/ anaphylaxis.

CLINICAL CORRELATION

Extra caution should be exercised when prescribing first-generation antihistamines to elderly people. The sedative effect can increase the risk of falls and confusion!

Motion sickness - First-generation antihistamines (e.g., promethazine and dimenhydrinate) have antimuscarinic properties which can be effective in treating motion sickness.

Nausea/ Vomiting - Some of the first-generation antihistamines, namely promethazine and chlorpheniramine, are useful in the treatment of nausea and/or vomiting. The mechanism is related to the antimuscarinic activity of these drugs.

Sleep aid - The lipophilic nature of the first-generation antihistamines allows them to easily cross the blood brain barrier and cause drowsiness. This effect of the drug is sometimes targeted to use them as non-FDA approved sleep aids.

CLINICAL CORRELATION

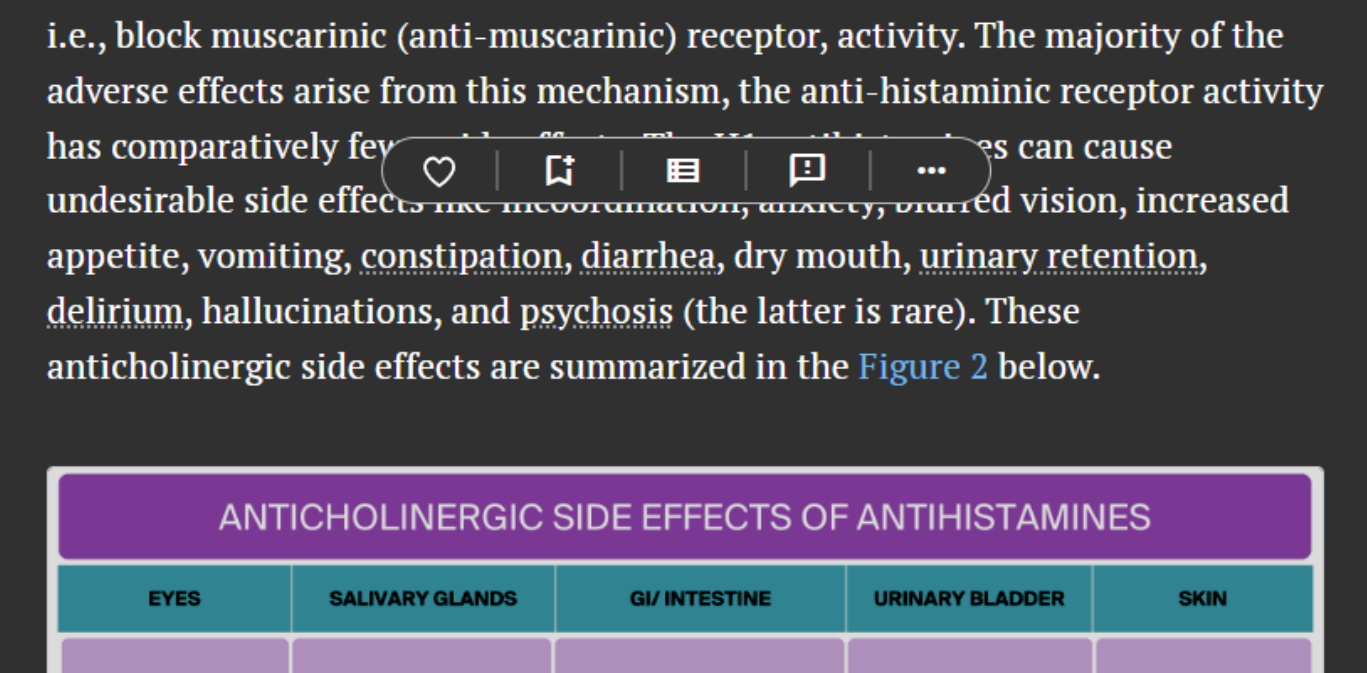
Nausea/ Vomiting in pregnancy - H1 antihistamines cross the placenta but are unlikely to be strong risk factors for major birth defects. The drug doxylamine, in combination with vitamin B6 (pyridoxine), is approved for the treatment of nausea and vomiting in preg

Why are first-generation antihistamines used as sleep aids?

What are the Adverse Effects of Antihistamines? (LO#4)

H1 Antihistamines

The first-generation antihistamines can cause unwanted **drowsiness**, as their more lipophilic nature, compared to second-generation antihistamines, allows them to readily cross the blood brain barrier. This can be particularly troublesome if the patient is taking other concomitant CNS depressants or ingests alcohol. Some first-generation H1 antihistamines have anticholinergic, i.e., block muscarinic (anti-muscarinic) receptor, activity. The majority of the adverse effects arise from this mechanism, the anti-histaminic receptor activity has comparatively fe s can cause undesirable side effects blurred vision, increased appetite, vomiting, constipation, diarrhea, dry mouth, urinary retention, delirium, hallucinations, and psychosis (the latter is rare). These anticholinergic side effects are summarized in the Figure 2 below.



ANTICHOLINERGIC SIDE EFFECTS OF ANTIHISTAMINES

EYES	SALIVARY GLANDS	GI INTESTINE	URINARY BLADDER	SKIN
Mydriasis (dilate), Decreased lacrimation	Decreased secretions (dryness)	Decreased GI secretions, Decreased vomiting, Decreased intestinal motility (constipation)	Decreased urination	Decreased sweating

Figure 2: Effects of Anticholinergic agents on various organs

H1 antihistamines readily cross the placenta, but they are unlikely to be strong risk factors for birth defects. However, due to the antimuscarinic properties of many first-generation H1 antihistamines they can reduce breast milk production in the lactating woman.

H2 Antihistamines

Second-generation a the side effects associated with the antimuscarinic activity. Though occasionally some patients may note dry mouth or mild drowsiness. The adverse effects associated with their antihistaminic activity is usually mild and non-specific. It may include headache, diarrhea, or fatigue.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to SY, what could be causing his new sleepiness?

You ask all your patients who are taking multiple medications to bring them to their appointments so you can be sure you know what they are taking. Among SY's many pill bottles is a nonprescription allergy pill that contains diphenhydramine. SY confirms that this is a new addition. Then he confirms that, yes, the sleepiness did start soon after he started taking this new pill. Mystery solved! You are relieved that helping SY could be as simple as stopping this medication and perhaps replacing it with something nonsedating like loratadine.

Summary

- Histamine mediates local immune responses, regulates physiological functions in the gastrointestinal tract, and acts as a neurotransmitter in the brain, spinal cord, and uterus.
- Histamine is stored in most body tissues, primarily in mast cells and basophils, and is often activated in response to allergens.
- H1 receptors are widespread throughout the body and targeted in medication therapy.
- H1 antihistamines 1 histamine receptors on cell surfaces, leading to effects like vasoconstriction and lessening symptoms like runny nose.
- First-generation H1 antihistamines can cross the blood-brain barrier and cause sedation, while second-generation antihistamines are less able to cross the blood-brain barrier and do not cause significant sedation.
- Antihistamines are commonly used to relieve symptoms from allergies, motion sickness, and non-FDA approved sleep aids.
- H1 antihistamines are associated with more adverse effects, such constipation, dry mouth, urinary retention, and delirium, due to their anti-muscarinic/ anticholinergic activity.

Review Questions

1. What is the primary effect of H1 antihistamines on endothelial cells?

A. ○ Increased edema formation
B. ● Vasoconstriction
C. ○ Vasodilation
D. ○ Histamine release
E. ○ Increased itching

Hide Explanation

The correct answer is vasoconstriction (B). H1 antihistamines bind to H1 receptors on endothelial cells, leading to vasoconstriction by inhibiting the release of nitric oxide and prostacyclin. Increased edema formation (A) is incorrect as H1 antihistamines reduce edema. Vasodilation (C) is the opposite effect of what H1 antihistamines cause. Histamine release (D) is not inhibited by H1 antihistamines; rather, their action is blocked. Increased itching (E) is incorrect as H1 antihistamines reduce itching.

2. Which of the following medications could be used to treat motion sickness?

A. ● Promethazine
B. ○ Cetirizine
C. ○ Fexofenadine
D. ○ Loratadine

Hide Explanation

The correct answer is promethazine (A). This is a first-generation antihistamine that readily minds to antimuscarinic receptors. It readily crosses the blood-brain barrier and binds to neurons, inducing drowsiness which can reduce motion sickness. In contrast, second-generation antihistamines like cetirizine (B), fexofenadine (C) and loratadine (D) are less likely to cross the blood-brain barrier and lack the antimuscarinic prope

3. Which H1 antihistamine would be non-sedating?

A. ○ Diphenhydramine
B. ○ Hydroxyzine
C. ● Loratadine
D. ○ Promethazine

Hide Explanation

The correct answer is loratadine (C). This is the only medication option from the list that is a second-generation non-sedating option. Diphenhydramine (A), hydroxyzine (B), and promethazine (D) are all first-generation H1 antihistamines that are more lipophilic and readily cross the blood brain barrier to centrally block histamine receptors and cause sedation/drowsiness.

4. Which side effect is common with first-generation antihistamines?

A. ● Dry mouth
B. ○ Increased mucosal secretions
C. ○ Insomnia
D. ○ Polyuria

Hide Explanation

First-generation antihistamines can cause a variety of side effects due to their anticholinergic activity, including dry mouth (A). Increased mucosal secretions (B) would be a result of cholinergic drugs and the opposite of antihistamines. First-generation antihistamines readily cross the blood brain, i.e. are lipophilic, and cause drowsiness rather than inability to sleep or insomnia (C). First-generation antihistamines most often cause urinary retention due to the antimuscarinic properties and do NOT cause Polyuria (D).

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanisms of action of antihistamines.~~
02 ☑ ~~Classif~~ ~~ations based on central vs. peripheral effects.~~

drowsiness rather than inability to sleep or insomnia (C). First-generation antihistamines most often cause urinary retention due to the antimuscarinic properties and do NOT cause Polyuria (D).

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanisms of action of antihistamines.~~
02 ☑ ~~Classify antihistamines into their respective generations based on central vs. peripheral effects.~~
03 ☑ ~~Discuss the clinical uses of antihistamines based on their pharmacologic effects.~~
04 ☑ ~~Differentiate the potential side effects of antihistamine drugs, based on their effects on histaminic versus muscarinic receptors.~~

Objective Confidence: 4/4